COURSE
OVERVIEW

RADIO FREQUENCY AWARENESS

The Radio Frequency Awareness course provides learners with an understanding of radio frequency radiation and the risks associated with working in RF environments. The course focuses on how RF radiation is generated, how it behaves, and how exposure levels are controlled.

Learners are introduced to electromagnetic radiation, RF safety zones, signage, and detection methods. Emphasis is placed on recognising RF hazards, understanding exclusion zones, and applying safe work practices to prevent overexposure.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Explain basic concepts related to radio frequency exposure
- Identify RF hazards and potential exposure risks
- Recognise RF safety zones and associated signage
- Describe RF exposure limits and exclusion zones
- Apply controls to avoid RF overexposure
- Follow site procedures when working in RF environments


INDEX

INTRODUCTION

GRAVITY VALUES

RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

INTEGRITY

Conducting day-to-day business with integrity at all times.

SOLUTIONS **DRIVEN**

Being solution driven.

QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

SITE ESTABLISHMENT

Before any work begins, the site must be safe.

Use this simple checklist

Follow the Rules: 01

Make sure you follow all client rules.

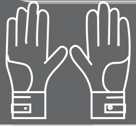
Example: If night work is not allowed, do not work at night.

Rescue Plan 02

Have a rescue plan ready. A rescue kit and first aid kit must be on site.

Work in Pairs 03

There must be at least two workers on site.
If something goes wrong, one can help or call for help.



Safe Site 05

Check the building or structure.

Make sure it is strong and safe to work on.

Check Equipment 04

All tools and gear must be safe and working before use.

06

Manage Traffic

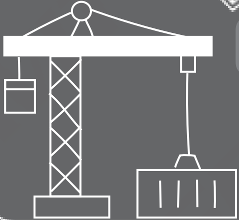
If there are cars or trucks nearby, there must be a traffic safety plan.

07

Do a Risk Assessment

Check what dangers there are.

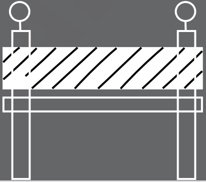
All workers must understand the risks before starting work.



08

Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

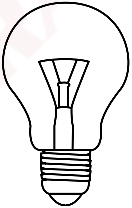
A. UNDERSTANDING RADIATION

1. What Is Radiation?

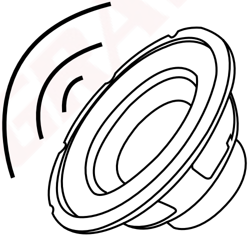
Any form of energy that moves through space is a form of radiation. There are four types of radiation:

- Alpha waves
- Beta waves
- Neutron waves
- Electromagnetic (EM) waves

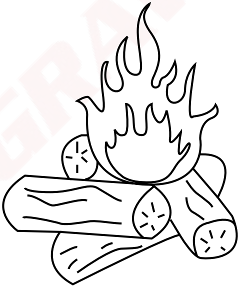
Let's look at some examples of different types of radiation we are exposed to daily:



Light radiation

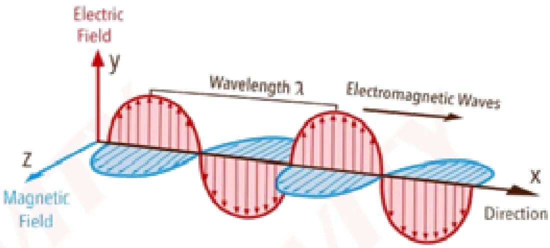


Acoustic radiation



Heat radiation

2. *What Is Electromagnetic Radiation?*



This course will be focussing specifically on electromagnetic radiation (EMR). EMR propagates (travels) in the form of an EM wave. An EM wave is created by a change in an electric field which creates a magnetic field and vice versa.

The frequency is determined by how many full cycles a wave can complete in one second. The amplitude is the value given to describe how strong a wave is (how much energy it carries).

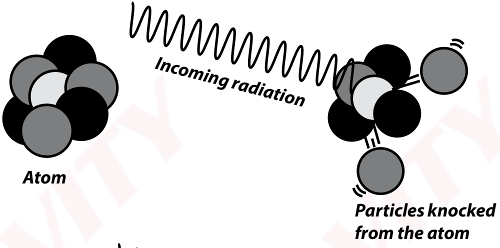
It is important to understand that the intensity of EM waves or radio frequency (RF) waves decreases as it propagates away from the source.

X-rays, light bulbs, the sun, fire, microwave ovens etc. all emit EM waves.

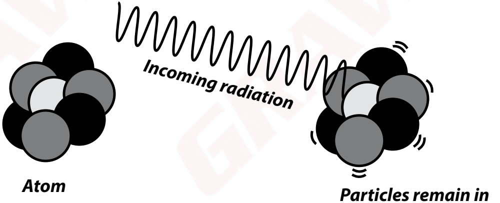
3. Ionising and Non-Ionising Radiation

Radiation is divided into two categories: ionising and non-ionising radiation.

If the incoming radiation from a source has enough energy, the radiation can **change the composition of an atom** by knocking out an electron. In other words, the radiation **ionises** the atom. This is referred to as **ionising** radiation. This type of radiation can lead to increased chances of diseases such as cancer.



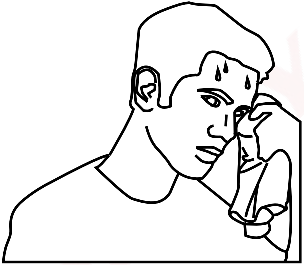
Now, if the incoming radiation does **not** have enough energy to remove particles from an atom, then the atom only **heats up**, thus **not ionising** the atom. This is referred to as **non-ionising** radiation (NIR). This can cause burns, heatstroke or organ failure, **but only at a very high intensity**. For low-intensity RF exposure, i.e. exposure below the International Commission on Non-Ionising Radiation Protection (ICNIRP) public limits, there is no health risk.



RF radiation is a form of NIR and it is important to understand the effects of exposure that this type of radiation can have on the human body and how to determine whether a person has been overexposed to radiation.

4. Effects of Non-Ionising Radiation

There are two main health effects that can result when exposed to high levels of RF energy:



Heating of the human body

Exposure to high levels of RF energy can cause a person to experience symptoms similar to sunburn or heatstroke.

Electromagnetic field (EMF) interference

EM radiation can convert any conductive material, such as metal implants or pacemakers, into antennas. This can lead to either heating or malfunctioning of devices.

B. ICNIRP AND RADIO FREQUENCY

1. ICNIRP

The **ICNIRP** provides scientific advice and guidance on the health and environmental effects of NIR to protect people and the environment from detrimental NIR exposure.

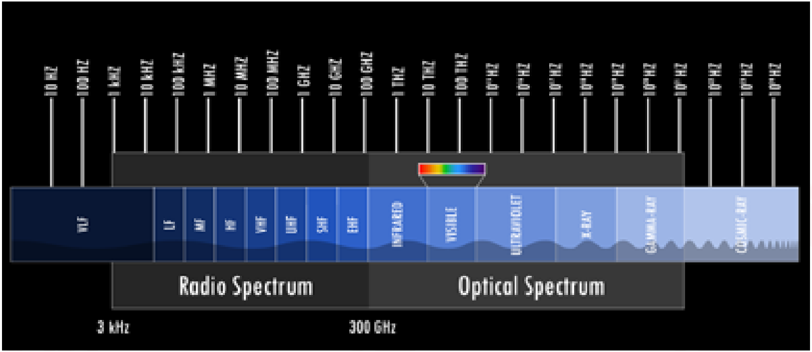
NIR refers to EM radiation such as ultraviolet, light, infrared and radio waves, and mechanical waves such as infra- and ultrasound. In daily life, common sources of NIR include the sun, household electrical appliances, mobile phones, Wi-Fi and microwave ovens.

Within the EM spectrum, NIR is situated below the ionising radiation band, which includes X-rays. NIR has less energy than ionising radiation and cannot remove electrons from atoms, i.e. NIR cannot ionise (except for part of the UV band). NIR is sub-grouped into different frequency or wavelength bands. The different subgroups have different effects on the body and require different protection measures.

The ICNIRP gives recommendations on limiting exposure for the frequencies in the different NIR subgroups. The ICNIRP develops and publishes guidelines, statements and reviews used by regional, national and international radiation protection bodies, such as the World Health Organisation. The ICNIRP is a main contributor to the international scientific NIR dialogue and the advancement of NIR protection.

ICNIRP website

2. The Electromagnetic Spectrum



Radio frequencies range from 3 kHz to 300 GHz and, as such, are invisible to the human eye, but this does not mean that they do not affect us.

(The ICNIRP guidelines are for the protection of humans exposed to RF electromagnetic fields (EMFs) in the range of 100 kHz to 300 GHz.)

Most of our current technologies operate in the RF part of the spectrum.

Sources of RF radiation include:

- Laptops
- Electric razors
- LED/LCD monitors

3. How to Avoid Exposure to RF Radiation

In order for workers to avoid RF overexposure, they must adhere to the following:

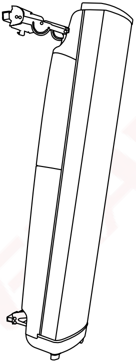
1. Move away from the RF source.
2. Switch off the RF source before commencing work.
3. Adhere to the ICNIRP guidelines.

To better understand how we can avoid overexposure, we need to understand how antennas work.

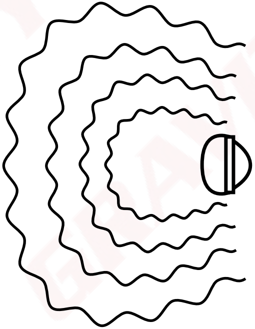
4. Directional Antennas

As the name suggests, directional antennas are directional, meaning they transmit into a single direction or within a specified range.

The easiest way to visualise this is to imagine using a flashlight in the dark. Though the light may light up the surrounding area, the majority of the light is focused in the direction you are pointing.

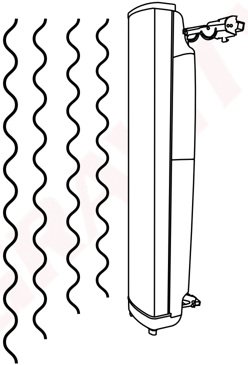


Directional antenna



Top view

RF radiation propagates out over a larger area, although it remains <u>in front</u> of the antenna.



Side view

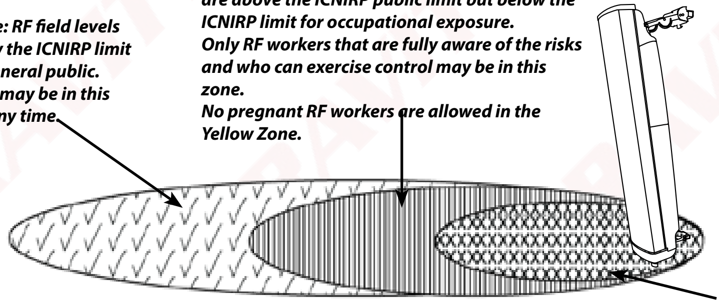
RF waves propagate outward away from the antenna and the intensity of the waves become less and less.

i. RF Safety Zones

Due to the dissipation of EM waves, there are different safety zones based on the acceptable levels of exposure.

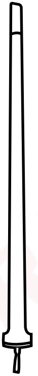
Three zones are classified based on the levels of RF exposure:

- Blue Zone
- Yellow Zone
- Red Zone

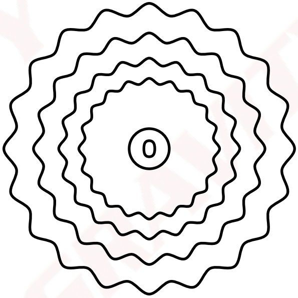


5. Omnidirectional Antennas

Contrary to directional antennas, omnidirectional antennas transmit horizontally in all directions. The easiest way to visualise this is to imagine using a ceiling light in your house. It lights up the entire area, but in terms of range, it is not as effective as a flashlight.

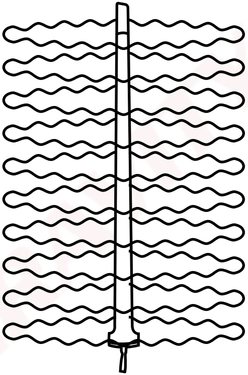


Omnidirectional antenna



Top view

If we were able to observe the transmission from an omnidirectional antenna, we would see that a worker will be exposed to RF radiation no matter on which side of the antenna they stand.



Side view

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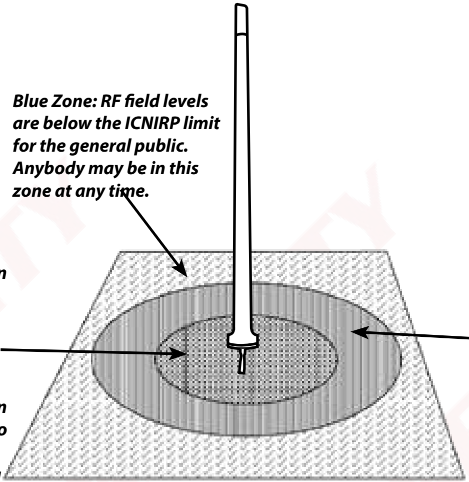
i. RF Safety Zones

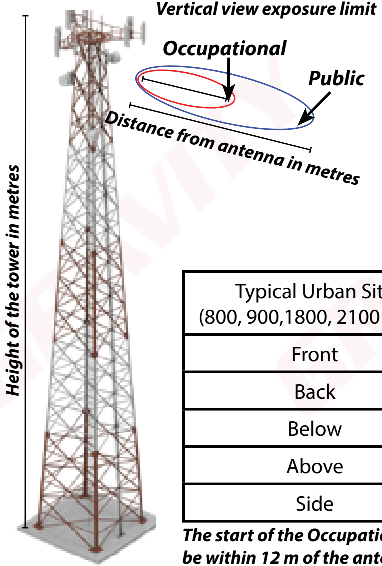


In real-world scenarios, we often hear workers say, "Since the installation was done, I have had constant headaches". But when installations are done, the exclusion zones are set up to ensure that people are not overexposed to radiation.

Depending on the type of antenna being installed, this is what you can typically expect to see in terms of how far the Public and Occupational Zones extend for rural and urban sites:

6. Rural Sites and Urban Sites

EMF distances from antenna



Typical Rural Site (800, 900 MHz)	Public Zone starts at	Occupational Zone starts at
Front	9.91 m	4.43 m
Back	28 cm	13 cm
Below	57 cm	26 cm
Above	46 cm	21 cm
Side	3.78 m	1.69 m

The start of the Occupational Zone is well out of reach of the public, as they would need to be within 10 m of the antenna to come near the outer limit of the Occupational Zone.

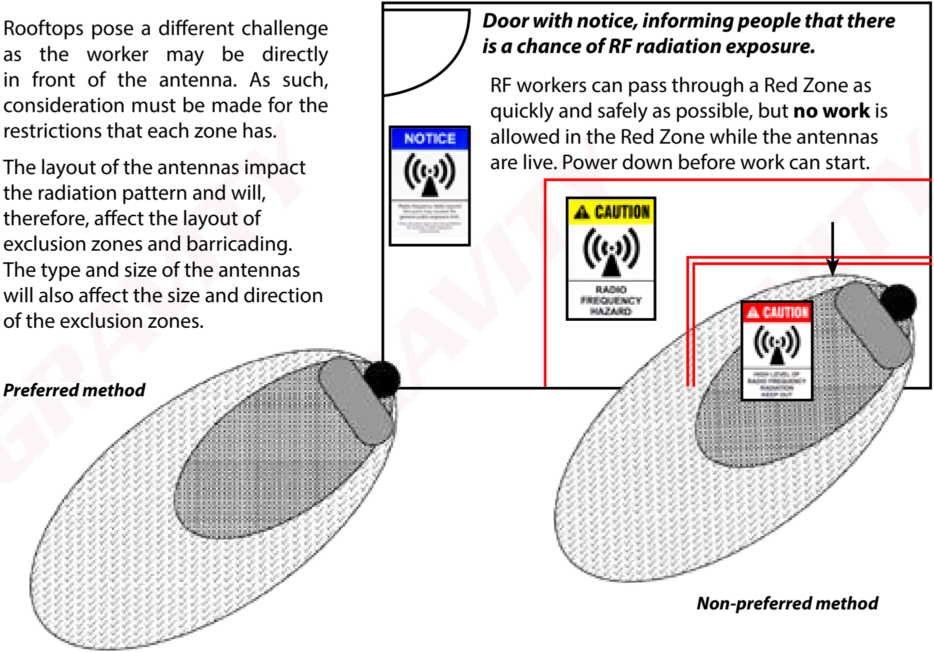
Typical Urban Site (800, 900,1800, 2100 MHz)	Public Zone starts at	Occupational Zone starts at
Front	11,49 m	5.14 m
Back	30 cm	13 cm
Below	76 cm	34 cm
Above	73 cm	32 cm
Side	4.59 m	2.05 m

The start of the Occupational Zone is well out of reach of the public, as they would need to be within 12 m of the antenna to come near the outer limit of the Occupational Zone.

7. RF Safety Zones on Rooftops

Rooftops pose a different challenge as the worker may be directly in front of the antenna. As such, consideration must be made for the restrictions that each zone has.

The layout of the antennas impact the radiation pattern and will, therefore, affect the layout of exclusion zones and barricading. The type and size of the antennas will also affect the size and direction of the exclusion zones.



Thus, the shape and size of the exclusion zones of an antenna are determined by:

- the type of antenna (omnidirectional, directional panel or beam forming massive multiple-input and multiple-output (MIMO)),
- the power transmitted by the antenna and
- multiple antenna installations close to each other.

The previous illustration is only an estimate of the safety zones of an average directional antenna. The frequency and power of the antenna will cause these zones to shift and change. **Therefore, it is of utmost importance to know which type of antenna you are working on.** See the manufacturer's instructions and recommendations in this regard.

Although these zones must be marked and easily visible, a good rule of thumb is to **never work directly in front of any RF antenna without following the relevant power-down procedures (as provided by the service provider as part of the safe working procedure)**.

In the case of a high-RF exposure zone, RF workers can pass through the Red Zone as quickly and safely as possible, but **no work** is allowed in the Red Zone while the antenna is live. Power down before work can start.

C. DETECTION METHODS

Since the output of antennas differ, the RF zones for different antennas will vary. This could cause workers to be unsure of where the Red Zone ends and the Yellow Zone begins. It is not possible to generalise RF zones for all antennas.

As such, you need to make use of dependable detection methods to identify the different zones and be sure where you may or may not work.

1. RF Monitors

One of the methods used to determine safe zones are RF monitors/scanners.
RF monitors are devices that help detect RF, and they aid in preventing accidental overexposure.
As the RF monitor is moved through an area, it measures the RF exposure at a specific location and for a specific time. It displays the results as a percentage of the ICNIRP occupational limits.

1. The RF monitor can be switched between "measure" and "monitor" mode:
- "Measure" is used to determine the strength of the field.
- "Monitor" is used to constantly ensure that the levels of exposure do not change.
2. The RF monitor has seven LED lights that light up one after the other in accordance with the measured intensity of the RF exposure. If four of the seven lights are lit, the radiation is approaching 100% of the exposure limit. Five lights means that exposure is exceeding 100%.
3. A good control measure is to avoid prolonged exposure in an area where the RF monitor starts beeping (indicating high RF levels).

fieldSENSE monitor

When using an RF monitor, always ensure that you:
- understand the operating modes and frequency bands,
- know which RF safety limits are referenced by the RF monitor and
- check that the RF monitor has a valid calibration date.

Any audio from the monitor indicates that the level of RF radiation is above the recommended exposure limit as set by the ICNIRP. In such cases, move away from the antenna or power down the antenna to continue working in that area.

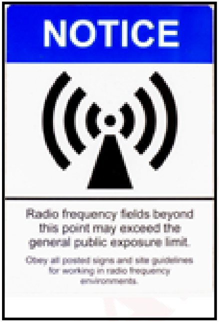
RF monitors can be very expensive and, as such, may not be available to everyone. On top of that, they are also battery operated, which may result in the monitor not working due to a depleted battery.

For this reason, most service providers and owners of properties where RF is present have used these monitors to determine the borders of different zones and have marked the zones with signage. Therefore, understanding what these signs mean is very important.

RF MONITOR INDICATORS

LEDs	Exposure Limit	Audio Alarm
7	250%	4 Hz
6	160%	2 Hz
5	100%	1 Hz
4	63%	1 Hz
3	40%	-
2	15%	-
1	5%	-

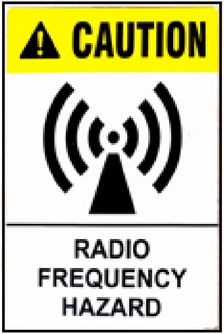
D. EMF SIGNAGE



1. The Three Zones

Blue Zone Sign

- This sign indicates that you are entering an RF controlled environment.
- Most sites use this sign at the gate as a precaution even if the levels on the ground are below the public safety limit.



Yellow Zone Sign

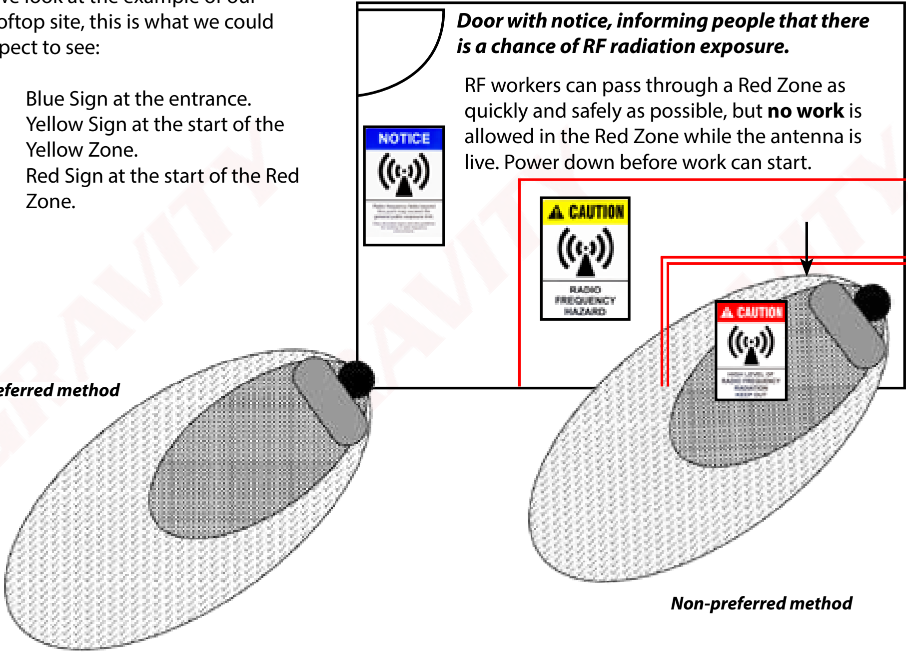
- This sign warns you that, beyond the sign, the RF fields may exceed the public safety limit.
- RF workers may go beyond the sign but the public may not.

Red Zone Sign

- This sign warns you that, beyond the sign, the RF fields exceed the occupational safety limit.
- RF workers are not allowed to work in the Red Zone.
- This sign should be used at any sites where there are areas on the ground at which RF exceeds the occupational safety limit.

If we look at the example of our rooftop site, this is what we could expect to see:

- Blue Sign at the entrance.
- Yellow Sign at the start of the Yellow Zone.
- Red Sign at the start of the Red Zone.



Preferred method

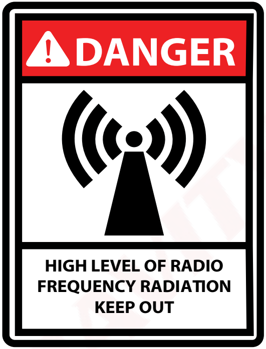
Non-preferred method

2. *Additional Caution/Danger Signs*

Other signs you could expect to see may warn you of dangerous radiation or hazards to electronic devices.

This sign represents **ionising** radiation and means that the area may not be entered by RF technicians without receiving the proper training.

This sign indicates that EMR can interfere with devices like pacemakers. These types of devices should not be exposed to any radiation.



- ***Beyond this point, RF fields exceed the occupational safety limit.***
- ***Nobody should enter the area unless power is reduced.***
- ***A Sentech site is an example of a high-power broadcasting site.***
- ***Use a suitable RF monitor to confirm exposure levels before working on high-power broadcasting sites.***

All these signs are of utmost importance in order to maintain **access control**. The risk assessment can be as thorough as possible, but it would be useless if there was no control over who enters the site and when. Exclusion zones prevent people from entering an area where they might be exposed to RF radiation.

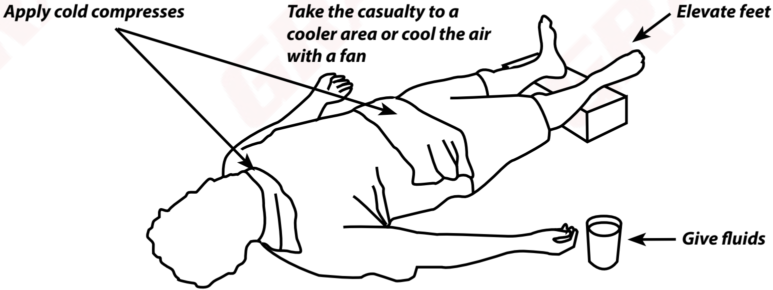
All telecommunication service providers have their own form of access control procedures which must be adhered to completely. When onsite, the safe work procedures given by the service provider must be adhered to at all times.

E. TREATMENT OF RF OVEREXPOSURE

We must always do our best to prevent any circumstance that may result in an incident, but in the event that something does go wrong, we need to know how to act.

Because RF exposure causes symptoms similar to that of heatstroke, our treatment for this type of "injury" is similar to how we would treat heatstroke:

- First Aider to move the casualty from the exposure area to a cool environment and provide cool drinking water.
- Apply cold water to burned areas.
- Seek immediate medical attention.
- Severe microwave or RF overexposure may damage internal tissues without apparent skin injury, so a follow-up physical examination is advisable.
- Report RF-related incidents to the site supervisor and to the service provider.



F. CONCLUSION

In order for workers to remain safe within an RF radiation environment, the following must be adhered to:

- Always follow the warning signs at the worksite.
- Try to remain out of the radiation zones of any antennas.
- Do not expose yourself to RF radiation for long periods of time.
- Should you be unsure of radiation levels, either evacuate the area or measure the levels and take the appropriate action.
- When needed, contact the RF safety officer for assistance.
- When RF workers get to the Red/Occupational Exclusion Zone of an antenna, they should pass through as quickly and safely as possible. No work is allowed in the Red Zone while the antenna is live.

G. THE WAY FORWARD

Congratulations on completing Gravity Training's Radio Frequency Awareness course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map, or consult your facilitator/assessor.

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